2019 EXAMINATIONS

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.212 Advanced Programming

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use a separate answer book for each question.

Question 1

Imagine you work for a computer games company as a software developer. You are working on a new Massively Multiplayer Online Role Play Game (MMORPG) called Gauntlet Online, where thousands of players collaborate and compete using virtual characters in an online fantasy world.

Your company has already made the decision to write Gauntlet Online using the Java language, and you have been placed in charge of designing the core Java classes that represent characters and items within the game. You have been given the following set of requirements:

- Every **character** has three core attributes: a unique user defined name, an indication of the health of the character (percentage value), and an inventory of **items** that player is carrying.
- Items may be given by one character to another, and they may have special abilities that can be activated by a player.
- An item's effect may be targeted on another character. For example, a player may use a sword on another player. When targeted on another character, that character is identified by name.
- The game must support any number of characters. There may be many thousands of players active at any point in time, each with their own character. There must be a java class to represent the game that provides an efficient way to find characters by name. Your colleagues have requested that, if possible, this be implemented using **static methods**, in order to simplify their code.

It is envisaged that player characters will be very long lived, gaining new abilities as the game progresses over months and years. Similarly, new types of items will be invented over time. Not all requirements are known at the time the product is launched, so **you should design your code to be extensible**. You will not be penalised for minor errors in syntax but should be as accurate as you can in your use of OO design principles.

Whilst paying particular attention to *OO design principles* throughout and writing *elegant*, *extensible*, and *type safe* code:

1.a Create appropriate Java classes to represent players and items within the game, taking care to model each of the attributes and capabilities listed in the requirements above.

[8 marks]

Question 1 continues next page…

Question 1 continued.

1.b Demonstrate how you might use your classes to create two characters – Gandalf and Balrog.

[2 marks]

1.c Demonstrate how you could write code to represent a **sword** held by Gandalf that can be used to inflict damage on a character.

[4 marks]

1.d A colleague wishes to add computer-controlled AI characters to the game. Show how you would create a **phantasm** AI character that, as soon as it is created, iterates through all the characters in the game and deals damage to their health.

[3 marks]

1.e Your colleague notices that your phantasm implementation affects other AI characters as well as player characters. Briefly describe how you might elegantly change your code such that a phantasm only targets player characters.

[3 marks]

Total 20 Marks

Question 2

2.a Consider the following list of programming principles and concepts. For each item in the list, define as precisely as you can what it means, describe the mechanism(s) in the Java language that implement it, and provide a simple example.

i. Inheritance **[3 marks]**
ii. Polymorphism **[3 marks]**
iii. Reflection **[3 marks]**
iv. Immutability **[3 marks]**

2.b The Java programming language allows the use of **generics**.

i. Describe what is meant by this term and the motivation for why generics are part of the Java language.

[4 marks]

ii. Write a Java code example that demonstrates how a simple generic class can be created and used. Create the simplest example you can that demonstrates the key concepts.

[4 marks]

Total 20 Marks

Question 3

3.a What is an 'Atom' in Erlang?

[1 mark]

3.b In JavaScript we can evaluate equality in two different ways, list both types and provide an example for each, demonstrating the difference.

[4 marks]

3.c All parts of this question refer to the following piece of Erlang code:

```
1  fizzbuzz( N, Result ) when N > 15 -> Result;
2  fizzbuzz( N, Result ) when ( N rem 3 =:= 0 ) and ( N rem 5 =:= 0 )
       -> fizzbuzz( N+1, Result ++ ["FizzBuzz"] );
3  fizzbuzz( N, Result ) when ( N rem 3 =:= 0 )
       -> fizzbuzz( N+1, Result ++ ["Fizz"] );
4  fizzbuzz( N, Result ) when ( N rem 5 =:= 0 )
       -> fizzbuzz( N+1, Result ++ ["Buzz"] );
5  fizzbuzz( N, Result ) -> fizzbuzz( N+1, Result ++ [N] ).
6
7  fizzbuzz( 1, [] ).
```

Note that as some lines of code are quite long, they have been split across two lines in this paper.

i. Explain in detail what each line does (refer to each line by the line number given).

[12 marks]

ii. Write the expected output from running this program.

[3 marks]

Total 20 Marks

Question 4

4.a Explain why the following expression will not work in Erlang:

```
X = X + 1
```

[2 marks]

4.b

i. Copy the following snippet of JavaScript code to your answer book, and for each line, explain in detail what that code does. Include what you would expect to see on the terminal when the program completes.

```
const f = x => y => x + (y * 2);
const data = [ 1, 2, 3, 4 ];
const result = data.map( f(2) );
console.log( JSON.stringify(result) );
```

[8 marks]

ii. The first line of the code in 4.b (i) defines a special form of a function. What is this known as? Explain how this is particularly useful for processing incomplete data.

[3 marks]

Question 4 continues on next page…

Question 4 continued.

4.c Consider the data shown in the following array called nba_teams (below).

```
const nba_teams = [
    {
      "conference": "Eastern",
      "division": "Atlantic",
      "name": "Boston Celtics",
      "city": "Boston",
      "state": "MA",
      "stadium": "TD Garden"
    },
    {
      "conference": "Eastern",
      "division": "Atlantic",
      "name": "Brooklyn Nets",
      "city": "New York",
      "state": "NY",
      "stadium": "Barclays Center"
    },
    {
      "conference": "Eastern",
      "division": "Atlantic",
      "name": "New York Knicks",
      "city": "New York",
      "state": "NY",
      "stadium": "Madison Square Garden"
    },
    {
      "conference": "Western",
      "division": "Pacific",
      "name": "Sacramento Kings",
      "city": "Sacramento",
      "state": "CA",
      "stadium": "Golden 1 Center"
       }
];
```

Write a **one-line** expression in JavaScript that would return the correct results for each of the following queries:

i. An array of all teams in New York

ii. A count of all Atlantic teams.

iii. The total length of all the value strings added together. **[7 marks]**

Total 20 marks

---End of Paper---